

Predicting Healthy Life Expectancy

Introduction

Our research question is: How do key variables— happiness score, economy (GDP per capita), and perceptions of corruption— affect healthy life expectancy, and how can we use statistical modeling to predict life expectancy based on these variables?

The dataset used in this study is sourced from Kaggle and titled, “[World Happiness Report \(2020-2024\)](#)” by Samith Chimminiyan. The data set contains 143 observations with variables including happiness score, economy (GDP per capita), social support, and others. The data is based on results from the Gallup World Poll, which contains internationally collected data based on several countries worldwide. In this study, we will focus on the 2024 data, and the variables “Happiness Score” (score of a country’s happiness), “Economy” (GDP per capita of a country), “Perceptions of Corruption” (score of a country’s perception of corruption), and “Healthy Life Expectancy” (score of a country’s life expectancy), using life expectancy as the response variable. These variables were selected out of all available variables in the data set because they represent key social and economic factors that could influence a population’s overall health. By analyzing the relationships among these variables, we can better understand their impact on healthy life expectancy and explore their interactions and contributions to variations in a country’s life expectancy.

To model the relationship between life expectancy and the selected predictors, we will apply multiple linear regression due to the multivariable data set we are using. Furthermore, we will explore transformations of this model to potentially improve statistical assumptions and significance in order to find the best predictive model.

This paper first explores the summary statistics, variable distributions, and the relationships between our response and predictor variables. It then explores and analyzes our untransformed full model, our log-transformed model, and our box-cox transformed model. It then discusses the best fit model based on the statistical and diagnostic assessment of those models. Finally, it summarizes the key findings of our data set and discusses its real-world applications, the model’s limitations, and any possible improvements to the decided model.

Data Description

We first start by looking at the summary statistics of the dataset. *Table 1* summarizes each variable’s mean and standard deviation to show the impact or contribution of the explanatory variables on the response variable (Healthy life expectancy). We can see that the Happiness score has the highest mean and contributes the most to a country’s healthy life expectancy, while the lowest contribution is perceptions of corruption. The Happiness score has the highest standard deviation, meaning it is more spread out across countries.

variable	mean	standard deviation
Healthy life expectancy	0.5209	0.1649
Happiness score	5.531	1.1812
Economy (GDP per Capita)	1.379	0.4251
Perceptions of Corruption	0.15412	0.1262

Table 1. Variable means and standard deviations

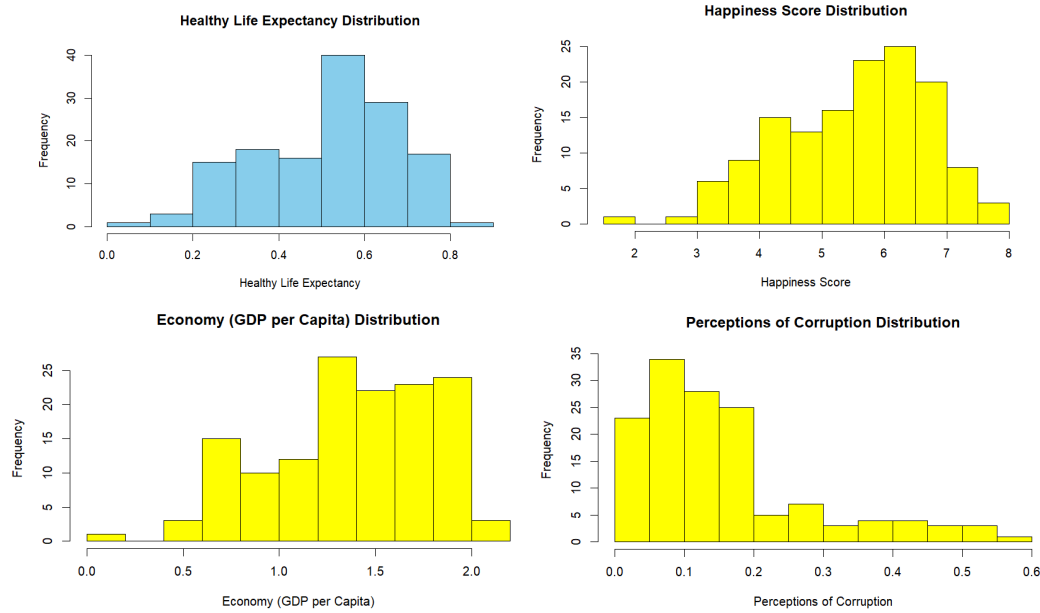


Figure 1. Distribution of variables

We can see that the Healthy life expectancy, Happiness, and Economy (GDP per Capita) scores are left skewed. The Perceptions of Corruption score is right-skewed.

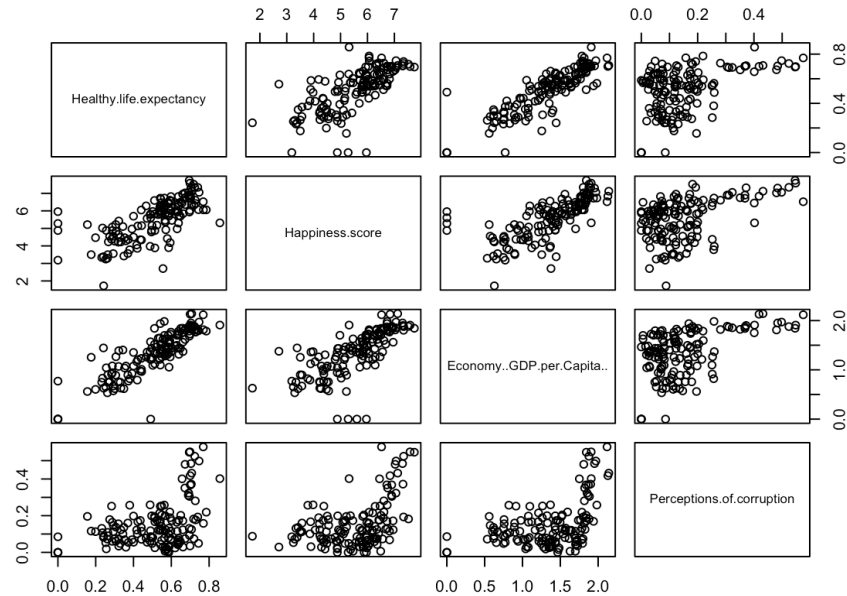


Figure 2. Scatterplot matrix of variables

Looking at the scatter plot matrix in Figure 2, we can see that Healthy life expectancy seems to have a positive linear relationship with Happiness score and Economy (GDP per Capita). Additionally, there appears to be homoscedasticity in the relationship. There seems to be no apparent relationship between Healthy life expectancy and Perceptions of corruption, which will be further investigated in a later analysis. Finally, there is no apparent relationship between the three predictors, suggesting random observations and no multicollinearity, which is what we want.

Since we observed that there is an apparent relationship between some of the explanatory variables and the response variable, we began analyzing by fitting our data with multiple linear regression models to determine the best model.

Results and Interpretation

Our first candidate model is the full model with untransformed data

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Call:
lm(formula = Healthy.life.expectancy ~ Economy..GDP.per.Capita.. +
    Perceptions.of.corruption + Happiness.score, data = happiness)

Residuals:
    Min       1Q   Median       3Q      Max
-0.28151 -0.04932  0.00136  0.04917  0.28909

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -0.030952   0.036357  -0.851   0.396
Economy..GDP.per.Capita..  0.233858   0.027645   8.459 3.75e-14 ***
Perceptions.of.corruption -0.004395   0.066769  -0.066   0.948
Happiness.score    0.041597   0.009991   4.163 5.53e-05 ***
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08736 on 136 degrees of freedom
Multiple R-squared:  0.7255,    Adjusted R-squared:  0.7194
F-statistic: 119.8 on 3 and 136 DF,  p-value: < 2.2e-16
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Figure 3. Summary Statistics for Candidate Model 1

From Figure 3 we derived the linear regression equation:

$$\widehat{\text{Healthy.Life.expectancy}} = -0.0310 + 0.234\text{Economy} - 0.00439\text{Corruption} + 0.0416\text{Happiness.score}$$

We can see that although Economy.GDP.per.Capita and Happiness.score have significant p-values for their coefficients, Perceptions.of.corruption does not, indicating that variable selection may be required later. In addition to the coefficients, the summary also reports an R^2 value of 0.7255, indicating that 72.55% of the variance in Healthy life expectancy is explained by the model. Finally, the summary shows that the overall F-test was significant with a p-value less than 0.05, rejecting the null hypothesis and giving significant evidence that at least one of the predictor coefficients is significant.

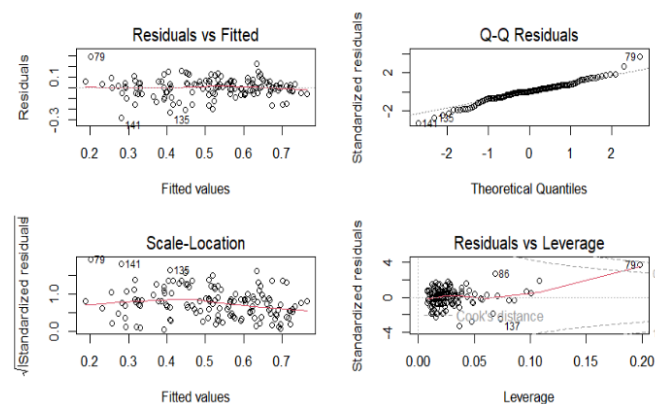


Figure 4. Diagnostic Plots for Candidate Model 1

Figure 4 suggests that the full model largely meets the assumptions of multiple linear regression. The Residuals vs. Fitted and Scale-Location plots indicate constant variance and mean-centered residuals,

supporting the assumptions of linearity and homoscedasticity. However, the Normal Q-Q plot reveals a slight right skew and heavy tails, suggesting some deviation from normality. Additionally, the Residuals vs. Leverage plot highlights a few outliers, specifically points 79, 86, and 137.

To determine whether a transformation could improve our diagnostic plots, we applied the Box-Cox method. The likelihood ratio test for a full log transformation yielded a p-value of 2.2×10^{-16} , which is significantly less than the significance level $\alpha = 0.05$, indicating that a complete log transformation could be appropriate. Similarly, the test for no transformation also produced a p-value of $2.2 \times 10^{-16} < \alpha$, suggesting that some form of transformation is necessary. Based on the Box-Cox recommendations, we rounded the suggested power transformations for each variable and refitted the multiple linear regression model using the transformed variables.

$$\widehat{(\text{Healthy.Life.expectancy})}^{1.43} = 0.102 + 0.106(\text{Economy})^2 - 0.059\text{Corruption} + 0.0030(\text{Happiness.Score})^2$$

In the second candidate model, all predictors were significant except for Perceptions.corruption. This model has an R^2 value of 0.7432, which is higher than that of the first candidate model. The F-statistic also yielded a significant p-value, indicating a strong linear relationship between the predictor and response variables in this model.

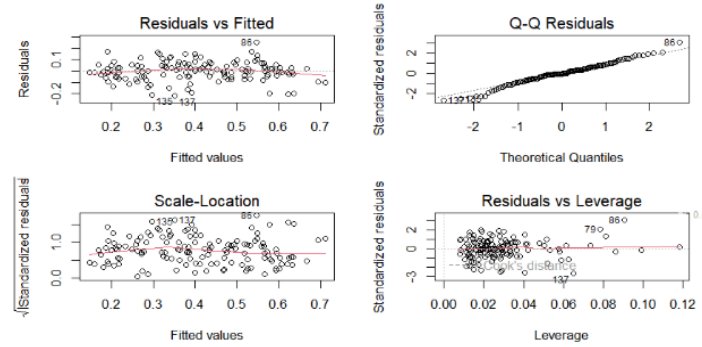


Figure 5. Diagnostic Plots of Candidate Model 2 (Transformed)

The diagnostic plots in Figure 5 show that many of the assumption violations still remain. The Residuals vs. Fitted and Scale-Location models still show that the trend line is horizontal around the mean, indicating linearity and constant variance assumptions. There is also a violation of the normality assumption, as the Normal Q-Q plot still has its tails bending. Finally, there also seems to be some leverage and influential points which can influence our model heavily, such as point 86 (could be a bad point)

Based on the summary of the multiple linear regression for Model 2, we investigated variable selection due to the high p-value of the Perceptions.of.corruption variable. We examined the added variable plots, in addition to finding the R^2 Adjusted, AIC, AIC corrected, and BIC for the model.

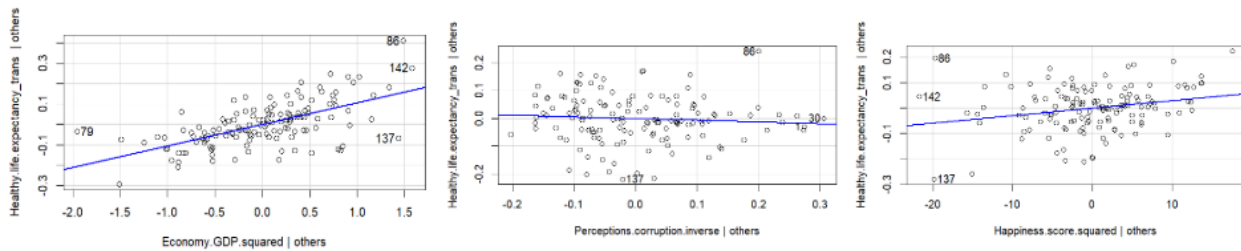


Figure 6. Added Variable Plot

Size	R^2 Adjusted	AIC	AIC Corrected	BIC
Economy..GDP.per.Capita.. (1)	0.6937	-672.9401	-672.8519	-155.6089
Economy..GDP.per.Capita..., Happiness.score (2)	0.7235	-686.1855	-686.0077	-165.9197
Economy..GDP.per.Capita..., Perceptions.of.corruption, Happiness.score (3)	0.7215	-684.1872	-683.8887	-160.9870

Table 2. R^2 Adjusted, AIC, AIC Corrected, BIC

The added variable plots from Figure 6 show us that both Economy.GDP.per.Capita and Happiness.score both have significant, positive slopes, and they have clustering around the trend lines, suggesting they have strong linear relationships and positive association with the response variable, while controlling for the other variables respectively. The predictor, Perceptions.of.corruption, is not significant since the slope is close to zero and there is a wider scatter of points around the trend line, this predictor and the response variable have little to no linear relationship. Looking at Table 2, we find consistent results seeing that the R^2 Adjusted, AIC, AIC Corrected, and BIC tests all indicate that the model with size two is the better model, consisting of the variables Economy.GDP.per.Capita and Happiness.score. Both predictors are significant with p-values less than 0.05, so they both have linear relationships with Healthy.life.expectancy. Also, the adjusted R^2 for the final model is high, explaining around 72% of the variance in Healthy.life.expectancy, which makes the model a good fit.

We arrive at our transformed and reduced final model:

$$(\widehat{\text{Healthy.Life.expectancy}})^{1.43} = -0.0087 + 0.235(\text{Economy})^2 + 0.038(\text{Happiness.score})^2$$

For our final model, we understand that the intercept being -0.0087 when both Economy.GDP.per.Capita and Happiness.score are zero represents that the power-transformed response variable, Healthy.Life.expectancy^{1.43}, is approximately zero. Then, we look at the Economy variable and Happiness.score, both of which are squared in the model, this indicates that the predictors are not linearly related to the response variable. Thus, we can say that the coefficients for Economy and Happiness.score, which are both positive, have positive relationships with Healthy.Life.expectancy, so when either predictor increases, healthy life expectancy also increases.

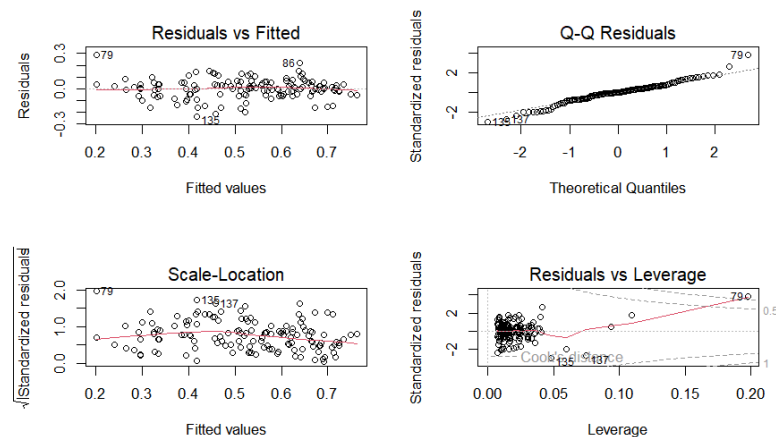


Figure 7. Diagnostic Plots of Candidate Model 3 (Final)

Taking a look at the diagnostic plot in *Figure 7* of our final model, we see that points 79, 135, and 137 are observations that may need further investigation. In our Residuals vs Fitted plot, we see slight clustering for the larger fitted values but the regression line is fairly flat and there is random scatter, so the linearity assumption holds. The Q-Q plot displays a diagonal line so the residuals are normal. There is deviation at the tails, slightly on the upper right end, due to point 79 which is an outlier. Thus, residuals are fairly normal, though there are some issues with outliers, but the normality assumption seems to hold. In the Scale-Location plot, there is a slight curve, which suggests heteroscedasticity, meaning there are some deviations as residuals increase with the fitted values. The observations 79, 135, and 137 stand out in particular, so the variance is not completely constant. The Residuals vs Leverage plot shows that point 79 has a high leverage and a high standardized residual, this indicates it might have a disproportionate influence on the regression. The other points are clustered at a lower leverage, so most observations are not highly influential. We see the transformation and reduced model did not fully address some assumption violations, with issues of influential points and in the variance. However, there are no major violations that invalidate the final model, and the model did improve the fit and the regression assumptions are reasonably met. We will keep the transformed and reduced model.

Discussion

Through an exploratory analysis of variables such as Happiness score, Economy (GDP per capita), and Perceptions of Corruption, this project aimed to identify potential influence on life expectancy. Our primary goal was to determine the best-fitting regression model, representative of the relationships between these factors and life expectancy. Our final model indicated that the Economy (GDP) and Happiness Score represent a positive relationship with Healthy Life Expectancy; with the Economy having a stronger influence (Coefficient: 0.233). Therefore, our results demonstrate significant predictive power between Economy and Happiness Score on the dependent variable of Healthy Life Expectancy.

In a broader context, these results suggest that countries with stronger economies (GDP per capita) and happier populations, will typically correlate with longer life expectancies. This model makes sense in a real-world situation as well. For example, studies such as *Happiness and Longevity in the United States*, emphasize the role of Happiness as a “meaningful subjective perspective on well-being that cannot be accounted for with more objective indicators of socioeconomic and social resources”¹. (Lawrence, Rogers, & Wadsworth, 2015) This suggests that even after controlling for demographic, socioeconomic, and lifestyle factors, happiness holds a distinctive relation to longer lives among U.S. adults. Economical conditions, measured by (GDP per capita) and life expectancy, have also been explored in current literature. Goran Miladinov’s paper on *Socioeconomic development and life expectancy relationship: evidence from the EU accession candidate countries*, investigated the relationship between socioeconomic development measured through GDP per capita, as an indicator of mortality or longevity. According to the findings, “higher values of GDP per capita and lower values of infant mortality levels lead to higher life expectancy at birth²,” suggesting that higher values of GDP per capita may be predictive of life expectancy.

Some limitations of our model include bias from omitted variables since our analyses involved the testing of three potentially predictive variables (Healthy life expectancy, Happiness score, Economy/GDP per Capita, and Perceptions of Corruption). While we determined to exclude Perceptions of Corruption as a predictor in our model due to its poor predictive power, other unrepresented variables in our model may increase the risk of our model underfitting. Additionally, while our dataset was composed of international data, the *World Happiness Report*³ does mention concerns about generalizability, considering there is a predominant emphasis on upper-middle-income countries when it comes to observations regarding ages 10-15. (Helliwell et al. 2024, 68). These limitations can be improved upon in the future by testing more variables for predictive influence on life expectancy, and by ensuring that our dataset does not have any limitations in generalizability.

Works Cited

- Helliwell, John F., et al. *World Happiness Report 2024*. Sustainable Development Solutions Network, 2024, <https://happiness-report.s3.amazonaws.com/2024/WHR+24.pdf>.
- Lawrence, Elizabeth M., et al. "Happiness and Longevity in the United States." *National Center for Biotechnology Information*, 2015, <https://pmc.ncbi.nlm.nih.gov/articles/PMC4724393>.
- Miladinov, Goran. "Does Happiness Matter? A Comparative Analysis of the Effect of Subjective Well-Being on Mortality in Europe." *Genus*, vol. 75, no. 1, 2019, <https://genus.springeropen.com/articles/10.1186/s41118-019-0071-0>.